Aggregated Datasets and Tools Inventory

Consolidated from 30 Replication Plans
REPLICATE Project

Auto-generated Aggregation

April 2026

Abstract

This document aggregates all datasets and computational tools referenced across the 30 replication plan `.tex` files in the REPLICATE project. Section 1 provides a consolidated master list of all datasets organized by type. Section 2 provides a deduplicated master inventory of all software tools, libraries, and frameworks organized by category. Section 3 provides a per-paper quick-reference summary.

Contents

Master Dataset Inventory

All datasets referenced across the 30 replication plans are consolidated below, organized by type.

Public Scientific Databases and Repositories

Dataset / Source	Paper(s)	Location / Notes
SDSS Stripe 82 light curves	2396968	https://www.sdss.org/dr17/
SDSS DR16Q quasar catalog	2396968	https://www.sdss.org/dr16/algorithms/qso_catalog/
ZTF light curves	2396968	https://www.ztf.caltech.edu/
Shen et al. (2011) BH mass estimates	2396968	Published catalog
LSST DESC photo-z challenge data	1461824	https://github.com/LSSTDESC/RAIL
SDSS photometric redshift catalog	1461824	https://www.sdss.org/dr17/
BPZ photo-z outputs	1461824	Via SDSS or DES data releases
Planck sky maps	2582579	https://pla.esac.esa.int/
Planck Sky Model (PSM)	2582579	https://pla.esac.esa.int/
UniProt / UniRef	1624105	https://www.uniprot.org/downloads
SCOP / SCOPe	1624105	https://scop.berkeley.edu/
Pfam (InterPro)	1624105, 2475938	https://www.ebi.ac.uk/interpro/
Metaclust	1624105	https://metaclust.mmseqs.com/
NCBI GenBank	2475938	https://www.ncbi.nlm.nih.gov/genbank/
NCBI SRA	2469515	https://www.ncbi.nlm.nih.gov/sra
IMG/VR database	2475938	https://img.jgi.doe.gov/vr/
CAMI synthetic community	2469515	https://data.cami-challenge.org/
PATRIC / BV-BRC genome database	2469515	https://www.bv-brc.org/
Materials Project (crystal structures)	1427646, 1981773	https://materialsproject.org/
Crystallography Open Database	1427646	https://www.crystallography.net/cod/
NIST Interatomic Potentials Repository	1609039	https://www.ctcms.nist.gov/potentials/
NIST Atomic Spectra Database	2441075	https://www.nist.gov/pml/atomic-spectra-database

Dataset / Source	Paper(s)	Location / Notes
CMIP5/CMIP6 ESM output	1578031	https://esgf-node.llnl.gov/search/cmip6/
Hector global-mean temperature	1578031	https://github.com/JGCRI/hector
MAGICC temperature pathways	1578031	https://magicc.org/
ENDF/B nuclear data libraries	2217719	https://www.nndc.bnl.gov/endf/
MSRE operational data	2217719	ORNL reports: ORNL-4449, ORNL-4396 via https://www.osti.gov/
NRC/EPRI CHF database	2571909	Contact NRC or EPRI for access
OEIS	1997354	https://oeis.org/
SNAP graph datasets	1379592	https://snap.stanford.edu/data/
SuiteSparse Matrix Collection	1379592	https://sparse.tamu.edu/
TU Dortmund graph benchmarks	2587225	https://chrsmrrs.github.io/datasets/
Parallel Workloads Archive (ANL, SDSC)	1984484	https://www.cs.huji.ac.il/labs/parallel/workload/
OSTI.gov abstracts	1861801	https://www.osti.gov/api/v1/
NRC ADAMS documents	1861801	https://www.nrc.gov/reading-rm/adams.html
DOE SciTech Connect	1861801	https://www.osti.gov/scitech/
dSprites dataset	2439897	https://github.com/deepmind/dsprites-dataset
HPOBench	2571540	https://github.com/automl/HPOBench
Experimental STEM images	1427646	From authors / ORNL data archives
Experimental SPM/STEM data	2439897	From authors / ORNL data repositories
PyEMMA alanine dipeptide dataset	1565592	Built into PyEMMA: http://www.emma-project.org/
DarkELF material data (Si, Ge)	3014512	https://github.com/tongylin/DarkELF

Synthetic / Generated Datasets

Dataset	Paper(s)	Notes
Paper inline examples (small matrices)	1379592	Extracted from paper text

Dataset	Paper(s)	Notes
Graph500 RMAT graphs	1379592	https://graph500.org/
Lattice geometries (triangular, kagome)	1412756	Generated from lattice vectors
Model parameters (J_K, J_H, t)	1412756	Specified in paper
Simulated STEM images (multislice)	1427646	Generated via PRISM / Dr. Probe
Mock photo-z PDFs (Gaussian mixtures)	1461824	Generated using `qp` utilities
Initial/boundary conditions (FDTD)	1475143	Specified analytically
Manufactured PDE solutions	1475143	Derived for convergence tests
Timing benchmarks (parallel runs)	1475143	Collected from target hardware
Rice–Mele / Haldane model parameters	1523841	From paper text
Band structure data	1523841	From Hamiltonian diagonalization
Synthetic Markov chain trajectories	1565592	Generated from specified transition matrices
Circuit parameters (qZSI)	1606674	From paper tables
Cu–Zr initial atomic configuration	1609039	Created by LAMMPS `create_atoms`
Synthetic flow fields (double gyre, vortex)	1842593	From analytic expressions
Synthetic trajectories (ODE integration)	1842593	From specified initial conditions
NN potential parameters (Argonne, Minnesota)	1864334	Published tables
VMC Monte Carlo samples	1864334	Generated by VMC code
CPW geometry parameters	1983793	From paper tables
Molecular integrals (PySCF)	1993311	Generated from quantum chemistry code
UEG parameters	1993311	From paper specification
DMQMC energy data	1993311	Output of QMC simulations
OEIS integer sequences (verification)	1997354	https://oeis.org/
CFI graphs	2587225	Standard construction
Random regular graphs	2587225	Generated with NetworkX
Simulated quasar light curves	2396968	From SDE models
Simulated microscopy images	2439897	Via multislice or analytical models

Dataset	Paper(s)	Notes
Benchmark test functions (Branin, Hartmann, etc.)	2571540	Standard definitions
DM halo model parameters	3014512	Standard Halo Model values
CAMB/CLASS C_ℓ spectra	2582579	Generated from Boltzmann solvers

Bundled / Repository Data

Dataset	Paper(s)	Notes
qp example/test data	1461824	https://github.com/aimalz/qp
fldgen test/example data	1578031	https://github.com/jgcri/fldgen
PyEMMA example datasets	1565592	http://www.emma-project.org/
SCALE example inputs	2217719	Distributed with SCALE package
CQSim test traces	1984484	Included in CQSim repository
SEEDtk tools and data	2469515	https://github.com/SEEDtk
ScaWL repository test graphs	2587225	From paper's code repository
NILC-PS-Model code + data	2582579	GitHub (see paper)

Master Tools Inventory

All software tools, libraries, and frameworks referenced across the 30 replication plans, deduplicated and organized by category.

Programming Languages

Language	Paper(s) (OSTI IDs)
Python 3.7+	1379592, 1412756, 1427646, 1461824, 1475143, 1523841, 1565592, 1609039, 1624105, 1842593, 1861801, 1864334, 1981773, 1983793, 1984484, 1993311, 1997354, 2217719, 2396968, 2439897, 2441075, 2469515, 2571540, 2571909, 2582579, 2587225, 3014512
R 4.0+	1578031
C/C++	1379592, 1475143, 1609039, 1624105, 2587225
Fortran	1412756, 1475143
Julia 1.6+	1412756, 1523841

Simulation and Domain-Specific Codes

Tool	Version	Paper(s)	Purpose / URL
LAMMPS	stable 2022+	1609039	Molecular dynamics `https://www.lammps.org/`
OVITO	$\geq$ 3.7	1609039	MD visualization `https://www.ovito.org/`
Atomsk	latest	1609039	Atomic structure manipulation `https://atomsk.univ-lille.fr/`
VASP	5.4+ / 6.x	1981773	DFT calculations `https://www.vasp.at/`
Quantum ESPRESSO	7.0+	1981773, 3014512	Open-source DFT `https://www.quantum-espresso.org/`
VESTA	$\geq$ 3.5	1981773	Structure visualization `https://jp-minerals.org/vesta/`
pymatgen	$\geq$ 2022.0	1981773	Structure manipulation `https://pymatgen.org/`
ASE	$\geq$ 3.22	1427646, 1981773	Atomic Simulation Env. `https://wiki.fysik.dtu.dk/ase/`

Tool	Version	Paper(s)	Purpose / URL
Bader analysis code	latest	1981773	Charge analysis http://theory.cm.utexas.edu/henkelman/code/bader/
VASPKIT	latest	1981773	VASP post-processing https://vaspkit.com/
sumo	latest	1981773	Band structure plots https://github.com/SMTG-Bham/sumo
SCALE	6.3+	2217719	Nuclear analysis suite https://www.ornl.gov/scale
OpenMC	latest	2217719	Open-source neutronics https://openmc.org/
Serpent 2	latest	2217719	Monte Carlo depletion http://montecarlo.vtt.fi/
HANDE	latest	1993311	QMC code https://github.com/hande-qmc/hande
PySCF	≥ 2.0	1993311	Quantum chemistry https://pyscf.org/
pyblock	latest	1993311	QMC error analysis https://github.com/jsspencer/pyblock
PRISM (Prismatic)	latest	1427646	STEM simulation https://prism-em.com/
abTEM	latest	1427646	Multislice simulator https://abtem.readthedocs.io/
Dr. Probe	latest	1427646	STEM simulation https://er-c.org/barthel/drprobe/
QuTiP	≥ 4.7	2441075	Quantum dynamics https://qutip.org/
ARC	latest	2441075	Atomic structure data https://arc-alkali-rydberg-calculator.readthedocs.io/
DarkELF	latest	3014512	DM–electron scattering https://github.com/tongylin/DarkELF

Tool	Version	Paper(s)	Purpose / URL
QEdark	latest	3014512	Alternative DM code `https://github.com/adrian-soto/QEdark`
PythTB	latest	1412756, 1523841	Tight-binding / Berry phase `https://www.physics.rutgers.edu/pythtb/`
kwant	≥ 1.4	1412756	Quantum transport `https://kwant-project.org/`
Z2Pack	latest	1523841	Wilson loop calculations `https://z2pack.greschd.ch/`
WannierTools	latest	1523841	Wannier Berry phase `https://www.wanniertools.org/`
MATLAB/Simulink	R2021b+	1606674	Circuit simulation `https://www.mathworks.com/`
PLECS	latest	1606674	Power electronics `https://www.plexim.com/plecs`
LTspice	latest	1606674	SPICE simulator
PySpice	latest	1606674	SPICE interface
COMSOL Multiphysics	6.0+	1983793	FEM EM simulation `https://www.comsol.com/`
Ansys HFSS	latest	1983793	FEM EM solver `https://www.ansys.com/`
Sonnet	latest	1983793	Planar EM simulator
PALACE	latest	1983793	SC circuit FEM `https://github.com/awslabs/palace`
Gmsh	≥ 4.10	1983793	Mesh generation `https://gmsh.info/`
FEniCS / FEniCSx	latest	1983793	Open-source FEM `https://fenicsproject.org/`
Qiskit Metal	latest	1983793	SC qubit design `https://qiskit.org/metal/`
NetKet	latest	1864334	VMC framework `https://www.netket.org/`

Tool	Version	Paper(s)	Purpose / URL
python-graphblas	latest	1379592	Python GraphBLAS API
SuiteSparse:GraphBLAS	≥ 7.0	1379592	C GraphBLAS implementation
CQSim	latest	1984484	Cluster queue simulator
CQGym	latest	1984484	OpenAI Gym wrapper

Machine Learning and Deep Learning Frameworks

Tool	Version	Paper(s)	Purpose
PyTorch	≥ 1.10	1427646, 1864334, 1984484, 2439897, 2571540, 2571909	Deep learning
JAX	latest	1864334, 2396968	JIT-compiled autodiff
torchvision	≥ 0.11	1427646	Backbones, augmentation
HuggingFace Transformers	≥ 4.15	1861801	Model loading, training
HuggingFace Datasets	latest	1861801	Dataset loading
RoBERTa (base/large)	pre-trained	1861801	Base LM
SciBERT	pre-trained	1861801	Scientific domain LM
seqeval	latest	1861801	NER evaluation
Stable-Baselines3	≥ 1.6	1984484	RL algorithms
OpenAI Gym / Gymnasium	≥ 0.26	1984484	RL environment
BoTorch	≥ 0.8	2571540	Bayesian optimization
GPyTorch	≥ 1.9	2571540, 2571909	GP modeling
Ax Platform	latest	2571540	High-level BO
scikit-optimize	latest	2571540	Alternative BO
Pyro	≥ 1.8	2571909	Probabilistic programming
AtomAI	latest	1427646, 2439897	DL for microscopy
e2cnn	latest	2439897	Equivariant CNNs
UMAP-learn	latest	2439897	UMAP visualization
PyEMMA	≥ 2.5	1565592	MSM estimation
deeptime	latest	1565592	MSM/Koopman analysis
celerite / celerite2	latest	2396968	Fast GP/SDE likelihood
GPy	latest	1993311	GP framework

Python Scientific Libraries

Library	Version	Paper(s)
NumPy	≥ 1.18	1379592, 1412756, 1427646, 1461824, 1475143, 1523841, 1565592, 1606674, 1609039, 1624105, 1842593, 1864334, 1981773, 1983793, 1984484, 1993311, 1997354, 2396968, 2439897, 2441075, 2571540, 2571909, 2582579, 3014512
SciPy	≥ 1.5	1379592, 1412756, 1427646, 1461824, 1475143, 1523841, 1565592, 1606674, 1609039, 1842593, 1864334, 1993311, 2396968, 2439897, 2441075, 2571540, 2571909, 2582579, 3014512
Matplotlib	≥ 3.3	All 30 papers (ubiquitous)
scikit-learn	≥ 0.24	1461824, 1565592, 1842593, 1861801, 1993311, 2439897, 2571909
scikit-image	≥ 0.18	1427646
Pandas	≥ 1.3	1984484, 2571909
SymPy	≥ 1.10	1997354, 2441075
NetworkX	≥ 2.6	1379592, 1997354, 2587225
Astropy	≥ 5.0	2396968
corner.py	latest	2396968
pathos	$\geq 0.2.8$	1461824
Seaborn	latest	1984484, 2571540, 2571909
Jupyter	latest	1461824, 2439897, 3014512
Numba or Cython	latest	1475143
pytest	≥ 6.0	1379592

R Packages

Package	Version	Paper(s)	Purpose
fldgen	v2.0	1578031	Field generation https://github.com/jgcri/fldgen
ncdf4	latest	1578031	NetCDF file I/O
vars	latest	1578031	VAR model fitting
raster / terra	latest	1578031	Geospatial data
ggplot2	latest	1578031	Visualization
Hector	latest	1578031	Simple climate model https://github.com/JGCRI/hector

MCMC and Bayesian Inference

Tool	Version	Paper(s)	Purpose
emcee	≥ 3.1	2396968, 2582579	MCMC sampler
PyMC	≥ 5.0	2396968	Bayesian modeling
cobaya	latest	2582579	Cosmological MCMC
GetDist	latest	2582579	MCMC chain analysis

Bioinformatics Tools

Tool	Version	Paper(s)	Purpose
MMseqs2 (Linclust)	latest	1624105	Protein clustering
CD-HIT	≥ 4.8	1624105	Protein clustering
USEARCH / UCLUST	latest	1624105	Protein clustering
DIAMOND	≥ 2.0	1624105	Protein alignment
MASH	latest	1624105	Genome distance
Prodigal	≥ 2.6	2475938	Gene prediction
HMMER	≥ 3.3	2475938	HMM protein search
MAFFT	≥ 7.4	2475938	Sequence alignment
trimAl	≥ 1.4	2475938	Alignment trimming
IQ-TREE	≥ 2.1	2475938	ML phylogenetics
RAxML-NG	latest	2475938	ML tree builder
BLAST+	≥ 2.12	2475938	Sequence similarity
ETE3	latest	2475938	Tree manipulation
iTOL	web	2475938	Tree visualization
CompareM / pyani	latest	2475938	AAI / ANI
BioPython	latest	2475938	Sequence manipulation
metaSPAdes	≥ 3.15	2469515	Metagenome assembly
RASTtk	latest	2469515	Genome annotation
PATRIC / BV-BRC	latest	2469515	Bioinformatics platform
SEEDtk	latest	2469515	Supervised binning
BWA-MEM	$\geq 0.7.17$	2469515	Read mapping
SAMtools	≥ 1.15	2469515	BAM/SAM manipulation
CheckM / CheckM2	latest	2469515	Genome quality
MetaBAT2	latest	2469515	Alternative binner
Trimmomatic / fastp	latest	2469515	Read trimming
MDTraj	≥ 1.9	1565592	MD trajectory I/O
OpenMM	≥ 7.5	1565592	MD simulation

Cosmology and Astrophysics Tools

Tool	Version	Paper(s)	Purpose
HEALPix / healpy	≥ 1.15	2582579	Spherical harmonics
CAMB	latest	2582579	CMB power spectra
CLASS	latest	2582579	Boltzmann solver
NaMaster (pymaster)	latest	2582579	Pseudo-C_ℓ estimation
NILC-PS-Model	latest	2582579	Paper's analysis code
`qp`	latest	1461824	Photo-z PDF approximation

Parallel Computing and HPC Libraries

Library	Version	Paper(s)	Purpose
MPI (OpenMPI / MPICH)	system	1475143, 2587225	Distributed parallelism
mpi4py	≥ 3.1	1475143	MPI Python bindings
OpenMP	system	2587225	Shared-memory parallelism
multiprocessing	stdlib	1475143	Python parallelism
LAPACK/BLAS	system	1412756	Matrix diagonalization
CMake	≥ 3.18	2587225	Build system
GCC	≥ 10	2587225	C/C++ compiler
GNU time	system	1624105	Benchmarking
nauty / Traces	latest	2587225	Graph isomorphism
ScaWL	latest	2587225	Scalable k-WL

Combinatorics and Symbolic Computation

Tool	Version	Paper(s)	Purpose
SageMath	≥ 9.5	1997354	Combinatorial enumeration
SymPy	≥ 1.10	1997354, 2441075	Symbolic computation
itertools	stdlib	1997354	Subset enumeration

Per-Paper Quick Reference

OSTI ID	Title (abbrev.)	Key Datasets	Key Tools
1379592	GraphBLAS Foundations	Paper examples, SNAP, SuiteSparse	Python, NumPy, SciPy, python-graphblas, NetworkX
1412756	Chiral Spin Order	Lattice geometries (synthetic)	Python/Julia, NumPy, SciPy, PythTB, kwant
1427646	Deep Learning STEM	Crystal structures, simulated & experimental STEM	PyTorch, PRISM, abTEM, Dr. Probe, ASE, AtomAI
1461824	Photo-z PDFs	Mock PDFs, qp test data, SDSS	Python, qp, NumPy, SciPy, scikit-learn
1475143	FDTD Delay PDE	Analytic IC/BC, manufactured solutions	Python, C/C++, NumPy, SciPy, mpi4py, MPI
1523841	Shift Photocurrent	Model parameters (synthetic)	Python/Julia, NumPy, PythTB, Z2Pack
1565592	Markov State Models	Synthetic chains, alanine dipeptide	Python, PyEMMA, deeptime, MDTraj, OpenMM
1578031	fldgen v2.0	fldgen data, CMIP5/6, Hector	R, fldgen, ncdf4, vars, ggplot2, Hector
1606674	CMV Reduction qZSI	Circuit parameters (paper)	MATLAB/Simulink, PLECS, LTspice, PySpice
1609039	$Cu_{64}Zr_{36}$ MD	Cu–Zr EAM potential (NIST)	LAMMPS, OVITO, Atomsk, Python
1624105	Linclust Clustering	UniProt, SCOP, Pfam, Metaclust	MMseqs2, CD-HIT, USEARCH, DIAMOND
1842593	Motion Tomography	Synthetic flow fields	Python, NumPy, SciPy, scikit-learn
1861801	NukeLM	OSTI, NRC ADAMS, DOE SciTech	HuggingFace, PyTorch, RoBERTa, SciBERT
1864334	VMC Neural Nuclei	NN potential params, reference energies	PyTorch/JAX, NumPy, SciPy, NetKet
1981773	Pt/LTO DFT	Crystal structures (MatProj)	VASP, QE, VESTA, pymatgen, Bader, sumo
1983793	CPW Resonator FEM	CPW geometry & material params	COMSOL, HFSS, PALACE, Gmsh, FEniCS
1984484	DRAS RL Scheduling	Parallel Workloads Archive	CQSim, CQGym, Stable-Baselines3, PyTorch
1993311	DMQMC + GPR	Molecular integrals, UEG params	HANDE, PySCF, scikit-learn, GPy, pyblock
1997354	Integer Sequences	OEIS sequences	SageMath, NetworkX, SymPy, itertools
2217719	SCALE MSR Depletion	ENDF/B, MSRE data	SCALE, OpenMC, Serpent 2, Python
2396968	Latent SDEs Quasars	SDSS Stripe 82, ZTF, BH masses	celerite2, emcee, PyMC, JAX, Astropy

OSTI ID	Title (abbrev.)	Key Datasets	Key Tools
2439897	Invariant VAEs	Microscopy images, dSprites	PyTorch, AtomAI, e2cnn, UMAP-learn
2441075	Light-Scattering Ions	NIST atomic data (Ba^{+})	QuTiP, SymPy, ARC, NumPy, SciPy
2469515	Metagenome Binning	NCBI SRA, CAMI, PATRIC	metaSPAdes, RASTtk, SEEDtk, CheckM
2475938	Virophage Taxonomy	GenBank, IMG/VR, Pfam HMMs	Prodigal, HMMER, MAFFT, IQ-TREE, BLAST+
2571540	Portfolio Bayesian Opt.	Benchmark functions, HPOBench	BoTorch, GPyTorch, PyTorch, Ax
2571909	Hybrid ML for CHF	NRC/EPRI CHF database	PyTorch, GPyTorch, Pyro, scikit-learn
2582579	NILC Cosmology	Planck maps, CAMB/CLASS spectra	HEALPix, CAMB, CLASS, NaMaster, emcee
2587225	ScaWL k-WL	CFI graphs, TU Dortmund	C/C++, MPI, OpenMP, nauty/Traces
3014512	Sub-GeV Dark Matter	DarkELF data (Si, Ge)	DarkELF, QEdark, Quantum ESPRESSO

Summary Statistics

Dataset Statistics

- **Total unique external databases/repositories:** 42
- **Papers with purely synthetic data:** 12 (1379592, 1412756, 1475143, 1523841, 1606674, 1842593, 1864334, 1983793, 1993311, 1997354, 2441075, 2571540)
- **Papers requiring public survey/catalog data:** 8 (1461824, 1565592, 1578031, 1624105, 1861801, 2396968, 2469515, 2582579)
- **Papers requiring licensed/restricted data:** 3 (2217719 – SCALE/ENDF, 2571909 – NRC CHF, 1981773 – VASP POTCARs)

Tool Statistics

- **Total unique tools/libraries identified:** ∼130
- **Most common language:** Python (27 of 30 papers, 90%)
- **Most common libraries:**
  - Matplotlib — 30/30 papers (100%)
  - NumPy — 24/30 papers (80%)
  - SciPy — 19/30 papers (63%)
  - scikit-learn — 7/30 papers (23%)
  - PyTorch — 6/30 papers (20%)
- **Papers requiring only open-source tools:** 24 (80%)
- **Papers requiring licensed software:** 6 (VASP, SCALE, MATLAB, COMSOL/HFSS, Serpent)

Domain Distribution

- Condensed matter / materials science: 6 papers
- Machine learning / deep learning: 6 papers
- Bioinformatics / genomics: 3 papers
- Astrophysics / cosmology: 3 papers
- Nuclear engineering / physics: 3 papers
- Computational physics / applied math: 4 papers
- Quantum computing / information: 2 papers
- Other (combinatorics, climate, power electronics): 3 papers